



MEDREACH

Medreach:

The right hospital, right now,
confirmed before you leave home

AI-powered emergency coordination for **Madhya Pradesh**



Understands **Hindi, English, Hinglish**



Hospital **accepts before you travel**



Live **patient tracking**

MPOnline Idea & Innovation Hackathon 2026 • Challenge 5, Healthcare

Live: medreach-wtne.onrender.com | github.com/prakharpurwal518-collab/Medreach



What Medreach Is

AI-powered emergency coordination, **not** a diagnosis app. It joins the family in panic, the hospital emergency desk and the health department on one live network.



CITIZENS & FAMILIES

Speak the emergency, see only hospitals that can treat it, get a confirmed bed and a tracking link. No login. "I'm alone" mode.



HOSPITALS

Verified board of beds, ICU, ventilators and doctors on duty. Referrals arrive with an AI pre-arrival (SBAR) note; one tap to accept.

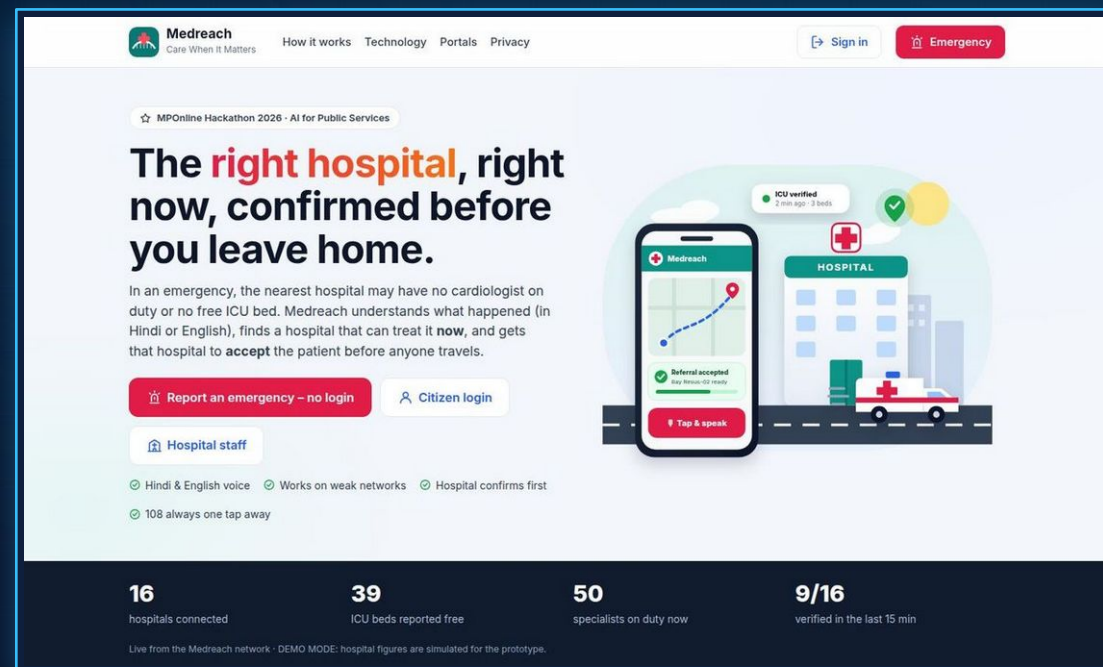


GOVERNMENT OF MP

A live, audited view of emergency capacity, and data on where patients are refused or delayed.

LIVE DEPLOYMENT

medreach-wtne.onrender.com



Our live home page, with network figures (demo data labelled as simulated).

7

suggested
technologies

2

private portals

16

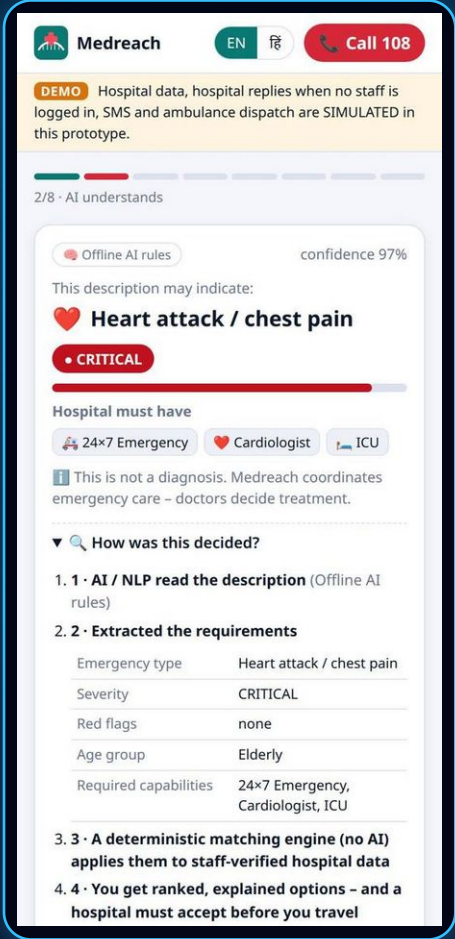
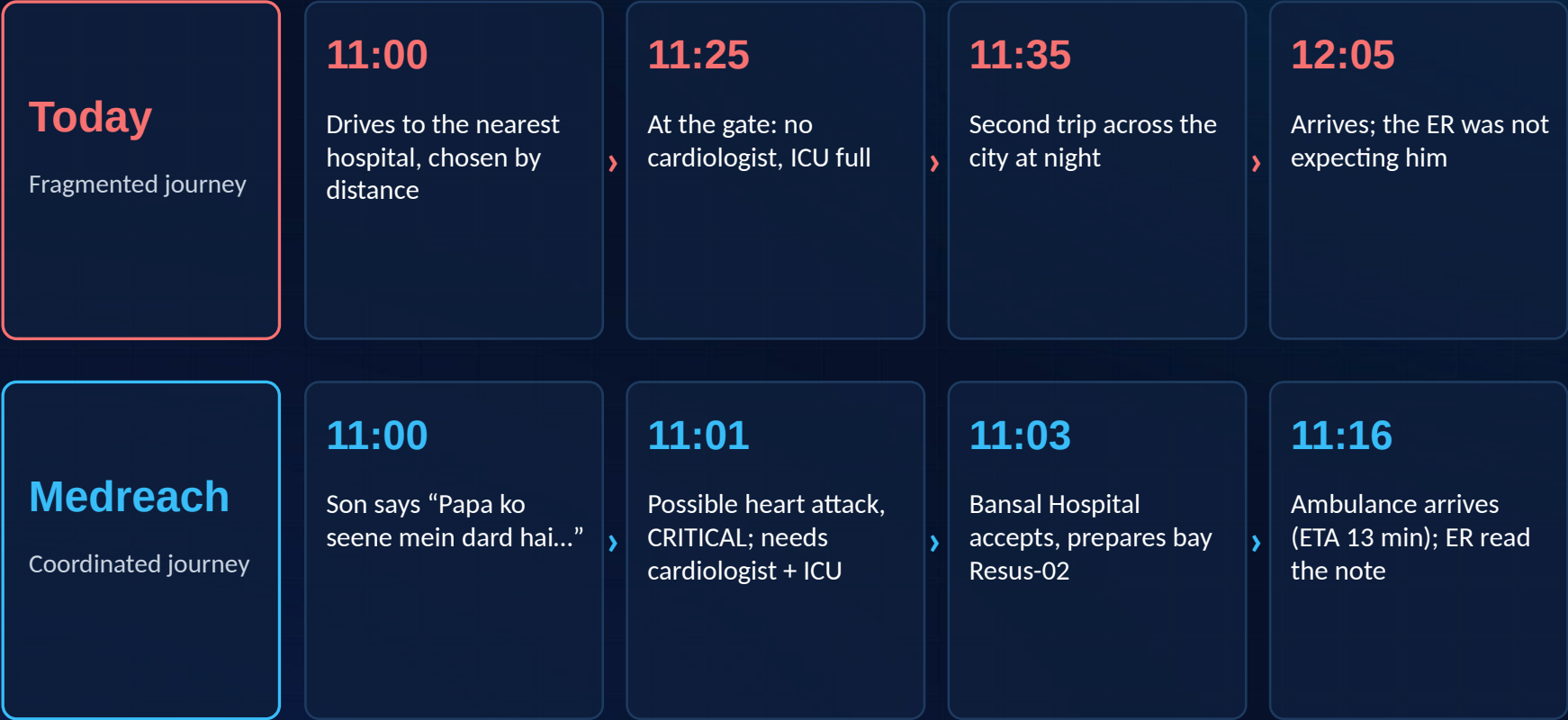
Bhopal-region
hospitals

51

automated tests

Chest Pain at 11 pm in Bhopal

A 62-year-old man has chest pain and is sweating. His son has a phone and a car.



Times are illustrative; the 13-minute ETA and bay come from our prototype run.

What the son sees a minute after speaking.

Why Patients Lose the Golden Hour



1,68,491

people died in road accidents in India in 2022

Source: MoRTH, "Road Accidents in India 2022". Heart attacks, strokes, burns and childbirth emergencies face the same delays.



Nearest is not capable

Maps show distance, not whether the specialist is on duty or the ICU has a bed tonight.



No trusted live status

Bed figures are old, with no owner or time stamp.



No acceptance first

Families arrive unannounced; refusal happens at the gate.



Panic and language

People cannot fill forms in panic; many prefer to speak Hindi.



People living alone

Cannot call hospitals, arrange transport and inform family at once.



Information is lost

Allergies and Ayushman / ABHA details don't reach the ER doctor.

WHO FEELS IT



Patients & families

wasted trips, delayed treatment



Rural citizens & ASHA workers

long distances, costly wrong choice



Hospital ER teams

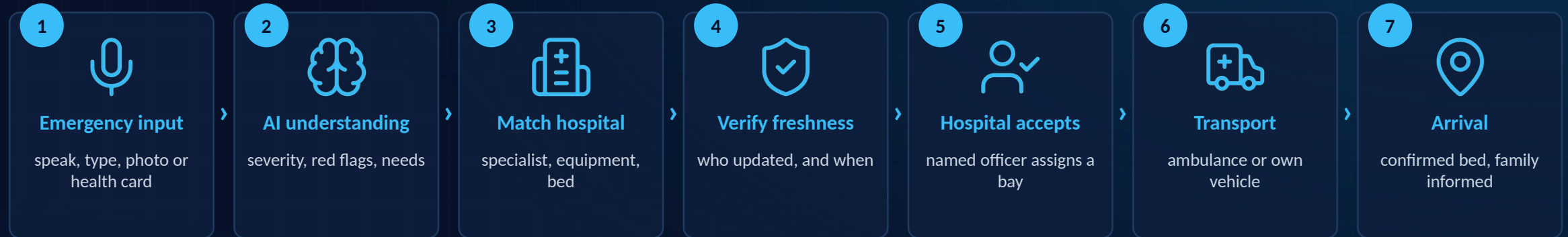
unannounced critical arrivals



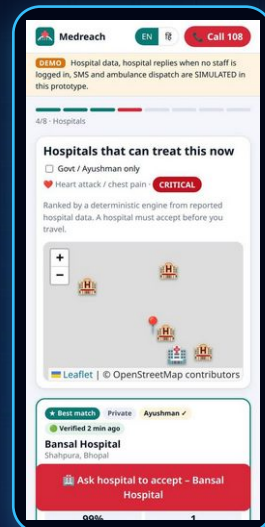
Health department

no live picture of capacity

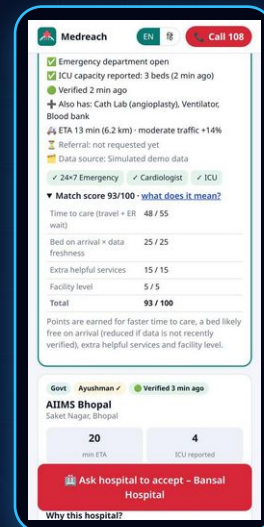
One Flow from Emergency to Confirmed Bed



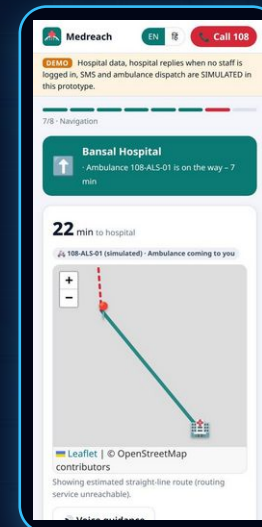
Speak or pick a common emergency



Only hospitals that can treat it



"Why this hospital?" score parts

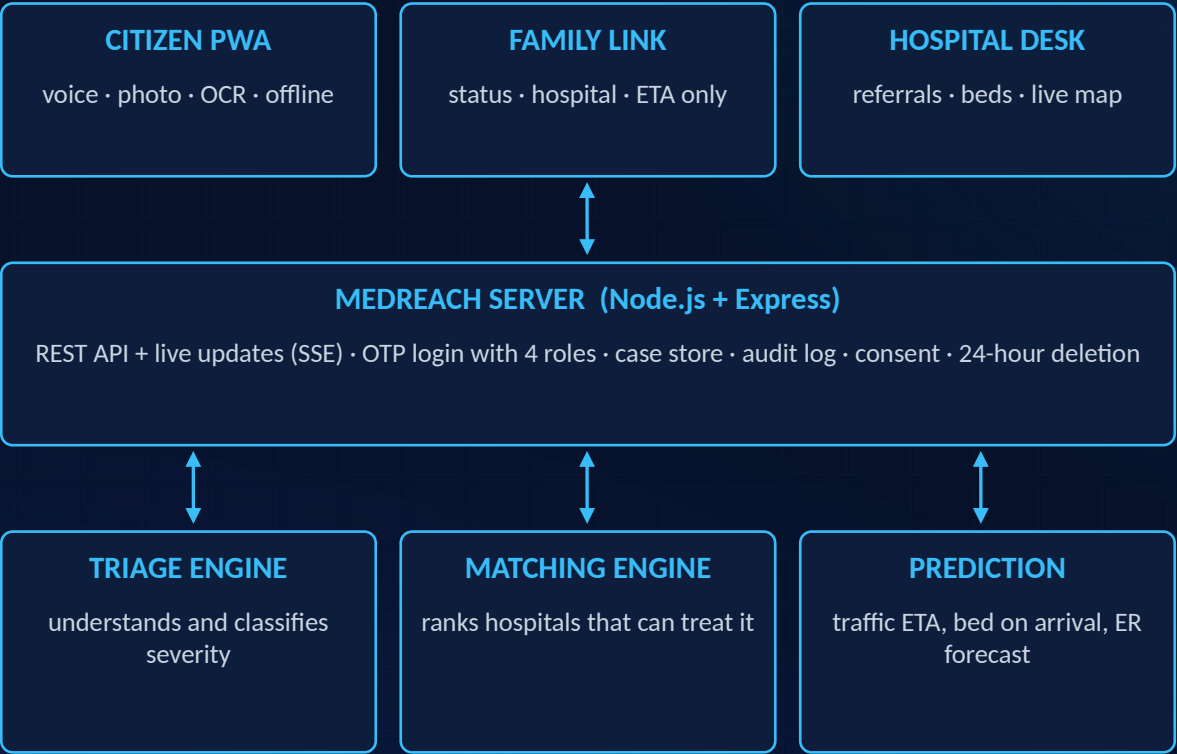


Ambulance status and navigation

RULES WE NEVER BREAK

- Hospital accepts before anyone travels
- Only hospitals that can treat the case
- 60 s without answer → next hospital
- Never diagnoses: "may indicate"
- Call 108 on every screen

Technical Approach



Engines run on the server and on the phone, so triage works offline. Claude (optional) adds understanding and writes the SBAR note.

Technology	What it does in Medreach
Generative AI	Turns spoken words into requirements; writes the ER note
Voice bots	Hindi / English speech in; key steps read aloud
Computer vision	Photo of an injury → visible findings
OCR	Reads Ayushman / ABHA cards on the phone, with consent
Predictive analytics	Traffic ETA, chance a bed is free on arrival, ER forecast
Conversational AI	One follow-up question at a time
Mobile app	Installable, bilingual PWA; works on weak networks

Stack: Node.js 22 · Express 5 · PWA · Leaflet + OpenStreetMap (no API keys) · Tesseract.js · Web Speech · Claude SDK

Safety: severity = max(AI, rules), so AI never lowers urgency. **Ranking:** time to care 55 + bed on arrival × freshness 25 + services 15 + level 5

From Prototype to Statewide



Prototype

Done

Portals live, AI triage, matching, acceptance, live maps, 51 tests



Pilot • Bhopal

~3 months

Hospital onboarding, nodal officers, ER training, clinical review, SMS OTP



Integrate

~3-6 months

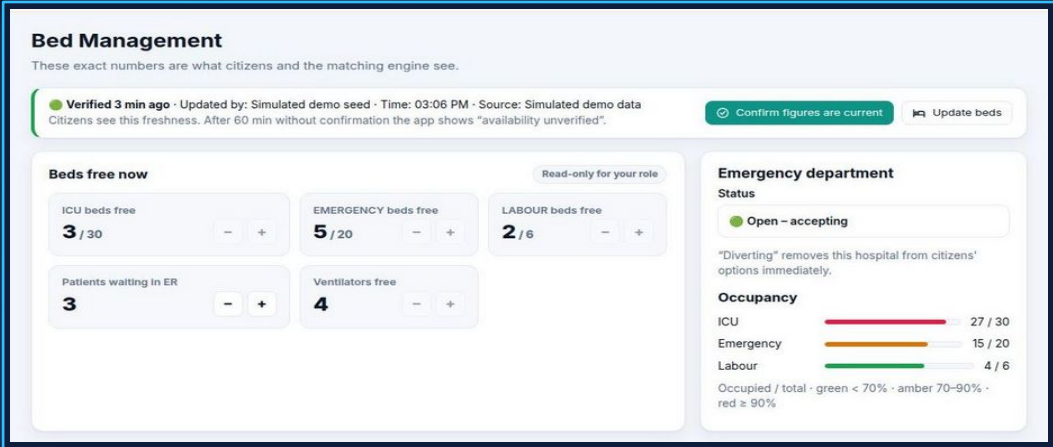
Health Facility Registry, HMIS, 108 control room, ABDM, encrypted DB



Scale • MP

~6-12 months

District rollouts, state command view, IVR for feature phones



What hospitals keep current. One click re-confirms the figures.

KEY RISKS & MITIGATIONS

Risk	Mitigation
Hospitals stop updating	Freshness on every figure; stale ranked lower; one-click confirm
AI misreads a case	AI only raises urgency; a deterministic engine decides
No hospital answers	After 60 s the next suitable hospital is asked
Weak network	Offline PWA runs triage and matching on the phone

What Makes Medreach Different

	Maps	Bed portals	Helplines	Medreach
Understands a spoken emergency in Hindi	No	No	Operator	Yes
Matches specialist and equipment	No	Beds only	Manual	Yes
Shows who updated each figure, and when	No	Rarely	No	Every figure
Hospital accepts before travel	No	No	Sometimes	Yes + bay
Automatic fallback to next hospital	No	No	Manual	60 s
Pre-arrival note for the ER doctor	No	No	No	SBAR note
Help for a person living alone	No	No	No	"I'm alone"

Reported capacity, referral accepted and confirmed destination are always shown separately, so “2 ICU beds” is never mistaken for “admitted”.



- 1
- 2

- 1 AI-written SBAR note from what the son said
- 2 Assign a bay and accept in one tap

< 1 s

status change reaches every citizen

60 s

before the next hospital is asked

0

API keys needed for maps

Impact and Benefits

01

Right care, first time

Pre-arrival acceptance gets patients treated sooner.

02

Equal access

Voice and Hindi for rural, elderly and first-time users.

03

Lower costs

No repeat trips; better use of ICU beds and specialists.

04

Live, audited data

Verified capacity for planning beds and ambulances.

05

Informed citizens

Safe first-aid steps; awareness of Ayushman, ABHA, 108.

06

Stronger system

Fewer surprise arrivals, less crowding, fewer transfers.

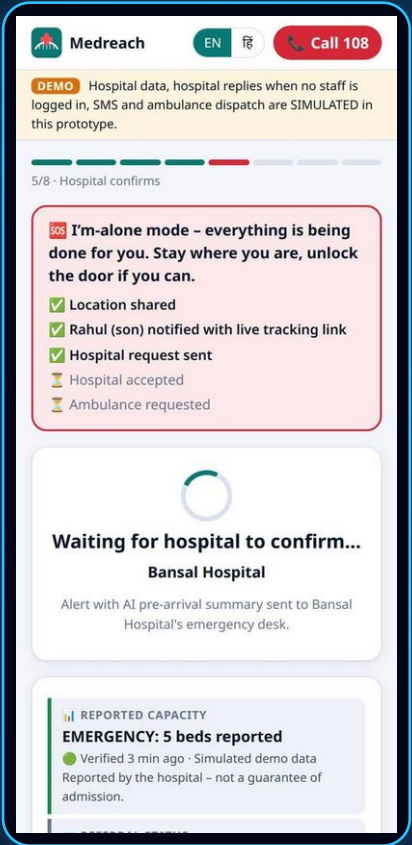
HOW THE PILOT WILL MEASURE IT

 Time to definitive care

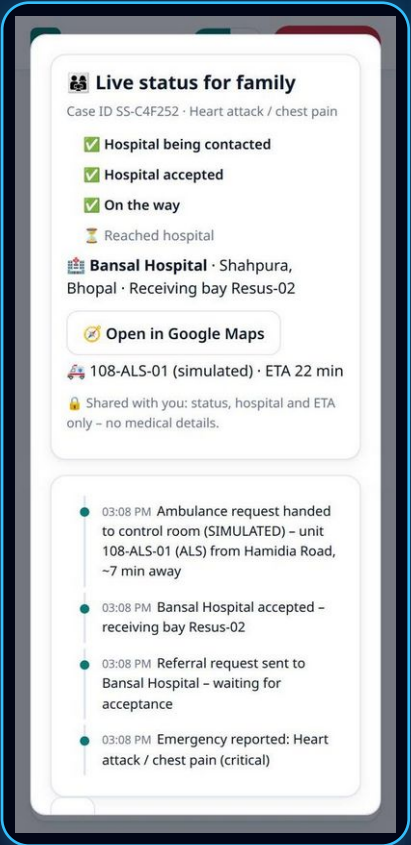
 Transfers avoided

 Acceptance time

 Data freshness



“I’m alone”: one tap does every step.



Family link: status, hospital, ETA. No medical data.

Why Should Medreach Be Selected?

Thank You



REAL IMPACT

Closes a life-or-death gap for citizens of Madhya Pradesh: a bed confirmed before travel.



WORKING NOW

Both portals are live on the web today. Try the whole flow in about 3 minutes.



BUILT IN FULL

All 7 suggested technologies in working code, listed file by file.



SAFE BY DESIGN

AI only raises urgency; a tested engine decides; 108 always one tap away.



PRIVATE

Consent, masked IDs, OTP and roles, audit log, deletion 24 h after hand-over.



READY TO SCALE

Low-cost open stack; clear path to HFR, HMIS, 108 and ABDM.

OUR ASK

Partner with us for a 3-month pilot with Bhopal hospitals, then statewide integration with the Health Facility Registry, hospital HMIS, the 108 control room and ABDM.

LIVE DEMO medreach-wtne.onrender.com

CODE github.com/prakharpurwal518-collab/Medreach

HOSPITAL LOGIN Bansal Hospital · BANSAL-ED01